

CEFEPIME-AFT

AFT Pharmaceuticals Ltd.

NAME OF THE MEDICINE

Cefepime-AFT Powder for Injection 1g

(cefepime hydrochloride monohydrate 1.189g equivalent to cefepime 1g)

DESCRIPTION

Cefepime for Injection is a sterile mixture of Cefepime Hydrochloride Monohydrate and Arginine. It is prepared by aseptically filling the sterile Cefepime Hydrochloride Monohydrate (Contains Arginine) into the vials.

Composition:

ACTIVE INGREDIENTS: Cefepime Hydrochloride Monohydrate 1.189g equivalent to 1g of Cefepime which is obtained from premix of cefepime hydrochloride monohydrate and L-Arginine.

INACTIVE INGREDIENTS: Arginine 720mg

Cefepime Hydrochloride is a β -lactam antibiotics and it is generally used to treat various infections. Arginine is added to control the pH of the constituted solution at 4.0~6.0.

The drug product is white to pale yellow powder. It is freely soluble in water.

The powder is filled into a colorless and transparent injection glass vials (type I glass) covered with rubber stoppers (primary packaging), and then sealed with aluminium cap, which is further protected in carton and big cardboard box (outer packaging).

The reconstituted solution is colorless to yellow solution, depending on the concentration and length of storage (for list of compatible reconstitution solution, see section DOSAGE AND ADMINISTRATION, Compatibility and Stability)

PHARMACOLOGY

Pharmacokinetics (adults)

Absorption

Average plasma concentrations of cefepime observed in normal adult males following single 30-minute intravenous infusions or intramuscular injection of 500mg, 1g and 2g are summarised in the Table 1. Following intramuscular administration, cefepime is completely absorbed.

Concentrations of cefepime achieved in specific tissues and body fluids are listed in the Table 2.

Table 1: Mean Plasma Concentrations of Cefepime (mcg/mL) in Healthy Adult Male Subjects

Cefepime dose	0.5 hour	1 hour	2 hours	4 hours	8 hours	12 hours
500mg, I.V.	38.2	21.6	11.6	5.0	1.4	0.2
1g, I.V.	78.7	44.5	24.3	10.5	2.4	0.6
2g, I.V.	163.1	85.8	44.8	19.2	3.9	1.1
500mg, I.M.	8.2	12.5	12.0	6.9	1.9	0.7
1g, I.M.	14.8	25.9	26.3	16.0	4.5	1.4
2g, I.M	36.1	49.9	51.3	31.5	8.7	2.3

Table 2: Mean Plasma Concentrations of Cefepime in Various Body Fluids (mcg/mL) and Tissues (mcg/g) in Healthy Adult Male Subjects

<i>Tissue or fluid</i>	<i>Dose (I.V.)</i>	<i>Average time of sample post-dose (hr)</i>	<i>Mean concentration</i>
Urine	500mg	0 - 4	292
	1g	0 - 4	926
	2g	0 - 4	3120
Bile	2g	9.4	17.8
Peritoneal fluid	2g	4.4	18.3
Blister fluid	2g	1.5	81.4
Bronchial mucosa	2g	4.8	24.1
Sputum	2g	4.0	7.4
Prostate	2g	1.0	31.5
Appendix	2g	5.7	5.2
Gallbladder	2g	8.9	11.9

Elimination

The average elimination half-life of cefepime is approximately 2 hours and the disposition of cefepime does not vary with respect to dose over the range 250mg to 2g. There is no evidence of accumulation in healthy subjects receiving doses of up to 2g intravenously every 8 hours for a period of 9 days. Total body clearance averages 120 mL/min. The average renal clearance of cefepime is 110 mL/min, demonstrating that cefepime is eliminated almost exclusively by renal mechanisms, primarily glomerular filtration.

Metabolism

Cefepime is metabolised to N-methylpyrrolidine which is rapidly converted to the N-oxide. Urinary recovery of unchanged cefepime represents approximately 85% of dose, resulting in high concentrations of cefepime in the urine. The serum protein binding of cefepime averages 16.4% and is independent of its concentration in the serum.

Special populations groups

Healthy volunteers aged 65 years old or older, who received a single 1g/vial I.V. dose of cefepime had higher AUC and lower renal clearance values compared to younger healthy adults. Dosage adjustments in the elderly recommended if renal function is compromised (see PRECAUTIONS and DOSAGE and ADMINISTRATION).

The pharmacokinetics of cefepime do not change to a clinically significant degree in cystic fibrosis patients. The pharmacokinetics of cefepime are unaltered in patients with impaired hepatic function who received a single 1g/vial dose. It is not necessary to alter the dosage of cefepime in these patient populations.

Studies in patients with varying degrees of renal insufficiency have indicated a prolongation in elimination half-life. There is a linear relationship between total body clearance and creatinine clearance in patients with abnormal renal function, which serves as the basis for dosage adjustment recommendations in this group of patients (see DOSAGE and ADMINISTRATION). The average half-life in severely impaired patients undergoing dialysis therapy is 13 hours for haemodialysis or 19 hours for continuous ambulatory peritoneal dialysis.

Pharmacokinetics (Paediatrics)

Single- and multiple-dose pharmacokinetics of cefepime have been evaluated in patients ranging in age from 2 months to 16 years who received 50 mg/kg doses administered by I.V. infusion; multiple doses were administered every 8 or 12 hours for at least 48 hours. Mean plasma concentrations of cefepime after the first dose were similar to those at steady state, with only slight accumulation seen upon repeated dosing.

Other pharmacokinetic parameters in infants and children were not different between first-dose and steady-state determinations, regardless of dosing schedule (q12h or q8h). There were also no differences in pharmacokinetics among the various patient ages or between male and female patients. Following a single I.V. dose, total body clearance averaged 3.3 mL/min/kg with the average volume of distribution being 0.3 L/kg. The overall mean elimination half-life was 1.7 hours. The urinary recovery of unchanged cefepime was 60.4% of the administered dose, and renal clearance was the primary pathway of elimination, averaging 2.0 mL/min/kg.

Microbiology

Cefepime hydrochloride for injection is a broad-spectrum cephalosporin antibiotic.

Cefepime is intended for intramuscular or intravenous administration.

Cefepime is a bactericidal agent that acts by inhibition of bacterial cell wall synthesis. It has a broad spectrum of activity against a wide range of gram-positive and gram-negative bacteria, including most strains resistant to aminoglycosides or third-generation cephalosporins. Cefepime is highly resistant to hydrolysis by most beta-lactamases, has a low affinity for chromosomally-encoded beta-lactamases, and exhibits rapid penetration into gram-negative bacterial cells. In studies using *Escherichia coli* and *Enterobacter cloacae*, cefepime bound with highest affinity to penicillin binding protein (PBP) 3 followed by PBP 2, then PBPs 1a and 1b. Binding to PBP 2 occurs with significantly higher affinity than that of other parenteral cephalosporins. This may enhance its antibacterial activity. The moderate affinity of cefepime for PBPs 1a and 1b probably also contributes to its overall bactericidal activity. Cefepime has been shown to be bactericidal by time-kill analysis (killing-curves) and by determination of minimum bactericidal concentrations (MBC) for a wide variety of bacteria. The cefepime MBC/MIC ratio was ≤ 2 for more than 80% of isolates of all gram-positive and gram-negative species tested. Synergy with aminoglycosides has been demonstrated in vitro, primarily with *Pseudomonas aeruginosa* isolates.

Cefepime has been shown to be active against most strains of the following organisms:

Gram-positive aerobes:

Staphylococcus aureus (including beta-lactamase-producing strains)

Staphylococcus epidermidis (including beta-lactamase-producing strains)

Other *staphylococci* including *S. hominis*, *S. saprophyticus*

Streptococcus pyogenes (Group A *streptococci*)

Streptococcus agalactiae (Group B *streptococci*)

Streptococcus pneumoniae (including intermediate penicillin resistant strains with penicillin MIC of 0.1 to 1 mcg/mL)

Other β -hemolytic *streptococci* (Groups C, G, F), *S. bovis* (Group D),

Viridans streptococci

Note: Most strains of *enterococci*, eg., *Enterococcus faecalis*, and *methicillin-resistant staphylococci* are resistant to most cephalosporins, including cefepime.

Gram-negative aerobes:

Pseudomonas sp. including *P. aeruginosa*, *P. putida*, *P. stutzeri*
Escherichia coli
Klebsiella sp. including *K. pneumoniae*, *K. oxytoca*, *K. ozaenae*
Enterobacter sp. including *E. cloacae*, *E. aerogenes*, *E. agglomerans*, *E. sakazakii*
Proteus sp. including *P. mirabilis*, *P. vulgaris*
Acinetobacter calcoaceticus (subsp. *anitratus*, *Iwoffii*)
Aeromonas hydrophila
Capnocytophaga sp
Citrobacter sp. including *C. diversus*, *C. freundii*
Campylobacter jejuni
Gardnerella vaginalis
Haemophilus ducreyi
Haemophilus influenzae (including beta-lactamase-producing strains)
Haemophilus parainfluenzae
Hafnia alvei
Legionella sp.
Morganella morganii
Moraxella catarrhalis (*Branhamella catarrhalis*) (including beta-lactamase-producing strains)
Neisseria gonorrhoeae (including beta-lactamase-producing strains)
Neisseria meningitidis
Providencia sp. including *P. rettgeri*, *P. stuartii*
Salmonella sp
Serratia including *S. marcescens*, *S. liquefaciens*
Shigella sp.
Yersinia enterocolitica

Note: Cefepime is inactive against many strains of *Stenotrophomonas maltophilia* (formerly known as *Xanthomonas maltophilia* and *Pseudomonas maltophilia*).

Anaerobes:

Bacteroides sp.
Clostridium perfringens
Fusobacterium sp.
Mobiluncus sp.
Peptostreptococcus sp.
Prevotella melaninogenica (formerly known as *Bacteroides melaninogenicus*)
Veillonella sp.

Note: Cefepime is inactive against *Bacteroides fragilis* and *Clostridium difficile*.

The prevalence of acquired resistance may vary geographically and with time for selected species. Information about the local resistance pattern should be obtained from a local bacteriological laboratory and taken into account in the choice of empiric therapy.

Susceptibility Tests

Diffusion techniques

Laboratory reports with standardized single disk susceptibility results using a 30 mcg cefepime disk, and in accordance with the National Committee for Clinical Laboratory Standards (NCCLS) should be interpreted according to the following criteria:

Zone Diameter (mm)	Interpretation
≥ 18	(S) Susceptible
15-17	(I) Intermediate
≤ 14	(R) Resistant

A report of “Susceptible” indicates that the pathogen is likely to be inhibited by generally achievable blood concentrations. A report of “Intermediate” indicates that the organism would be susceptible when high dosage is used or when the infection is confined to tissues and fluids (e.g., interstitial fluid and urine) in which high antibiotic levels are attained. A report of “Resistant” indicates that the achievable concentration of the antibiotic is unlikely to be inhibitory and other therapy should be selected. Organisms should be tested with the cefepime disk because cefepime has been shown to be active in vitro against certain strains found to be resistant with other beta-lactam disks. The cefepime disk should not be used for testing susceptibility to other cephalosporins. Standardized quality control procedures require the use of control organisms.

Dilution techniques

Using standardized dilution methods or an equivalent (eg., E-test™), the MIC values obtained should be interpreted according to the following criteria:

MIC (mcg/mL)	Interpretation
≤ 18	(S) Susceptible
16	(I) Intermediate
≥ 32	(R) Resistant

As with diffusion techniques, dilution techniques require the use of laboratory control organisms.

Insufficient data exist to support the efficacy of cefepime monotherapy in patients at high risk for severe infection (including patients with a history of recent bone marrow transplantation, with hypotension at presentation, with an underlying hematologic malignancy, or with severe or prolonged neutropenia). No data are available in patients with septic shock.

INDICATIONS

Adults:

Cefepime-AFT is indicated in the treatment of the infections listed below when caused by susceptible bacteria.

- Lower respiratory tract infections, including pneumonia and bronchitis.
- Urinary tract infections, both complicated, including pyelonephritis, and uncomplicated infections.
- Skin and skin structure infections.
- Intra-abdominal infections, including peritonitis and biliary tract infections.

- Gynaecological infections.
- Septicaemia
- Empiric treatment in febrile neutropenic patients (See PRECAUTIONS)

Paediatrics:

Cefepime-AFT is indicated in paediatric patients over 2 months of age for the treatment of the infections listed below when caused by susceptible bacteria:

- Pneumonia
- Urinary tract infections, both complicated, including pyelonephritis, and uncomplicated infections
- Skin and skin structure infections
- Septicaemia
- Empiric treatment in febrile neutropenic patients (See PRECAUTIONS)

Culture and susceptibility studies should be performed when appropriate to determine susceptibility of the causative organism(s) to cefepime. Empiric therapy with Cefepime-AFT may be instituted before results of susceptibility studies are known; however, once these results become available, the antibiotic treatment should be adjusted accordingly.

Because of its broad spectrum of bactericidal activity against gram-positive and gram-negative bacteria, Cefepime-AFT can be used appropriately as monotherapy prior to identification of the causative organisms(s). In patients who are at risk of mixed aerobic-anaerobic infection, particularly if bacteria not susceptible to cefepime may be present, concurrent initial therapy with anti-anaerobic agent is recommended before the causative organism(s) is(are) known.

DOSAGE AND ADMINISTRATION

Cefepime-AFT can be administered either intravenously or intramuscularly every 12 hours. The dosage and route vary according to the susceptibility of the causative organisms, the severity of the infection, renal function and overall condition of the patient.

Adults

Guidelines for dosage of Cefepime-AFT for adults are provided in Table 5.

Table 5: Recommended Dosage Schedule for Adults with Normal Renal Function (Aged 12 Years and Older)*

Severity of Infection	Dose & route of administration	Dosing Interval
Mild to moderate urinary tract infections	500mg – 1g/vial I.V. or I.M.	q12h
Mild to moderate infections other than UTI	1g/vial I.V. or I.M.	q12h
Severe infections	2g I.V.	q12h
Very severe or life-threatening infections	2g I.V.	q8h

*The usual duration of therapy is 7-10 days; however, more severe infections may require longer treatment. For empirical treatment of febrile neutropenia, the usual duration of therapy should not be less than 7 days or until resolution of neutropenia.

Surgical Prophylaxis

The dose recommendation for prophylaxis to prevent infection in adults undergoing intra-abdominal surgery is as follows:

- A single 2g I.V. dose of cefepime (as a 30-minute infusion, see DOSAGE AND ADMINISTRATION) starting 60 minutes before initial surgical incision. A single 500mg I.V. dose of metronidazole should be administered immediately following completion of the cefepime infusion. The metronidazole dose should be prepared and administered in accordance with official product labelling. Due to incompatibility, cefepime and metronidazole should not be mixed together in the same container (see COMPATIBILITY AND STABILITY); flushing of the intravenous line with a compatible fluid before infusion of the metronidazole is recommended.
- If the surgical procedure lasts longer than 12 hours from the initial prophylactic dose, a second dose of cefepime followed by metronidazole should be administered 12 hours following the initial prophylactic dose.

Paediatrics (aged 2 months up to 12 years with normal renal function)

Usual recommended dosages:

Pneumonia, urinary tract infections, and skin and skin structure infections: Patients > 2 months of age with body weight ≤ 40kg: 50 mg/kg q12h for 10 days. For more severe infections, a dosage schedule of q8h can be used.

Septicaemia and Empiric treatment of febrile neutropenia: Patients > 2 months of age with body weight ≤ 40kg: 50 mg/kg q8h for 7-10 days.

Experience with the use of cefepime in pediatric patients < 2 months of age is limited. While this experience has been attained using the 50 mg/kg dose, modeling of pharmacokinetic data obtained in patients > 2 months of age suggests that a dosage of 30 mg/kg q12h or q8h may be considered for patients aged 1 month up to 2 months. Both the 50 mg/kg dose for patients > 2 months of age and the 30mg/kg dose for patients between 1 and 2 months of age are comparable to a 2g adult dose. Administration of cefepime in these patients should be carefully monitored.

For pediatric patients with body weights > 40kg, adult dosing recommendations apply (see Table 5). For patients older than 12 years who are ≤ 40kg, the dosage recommendations for younger patients ≤ 40kg should be used. Dosage in pediatric patients should not exceed the maximum recommended dosage in adults (2g q8h). Experience with intramuscular administration in pediatric patients is limited.

Patients with Impaired Renal Function

In patients with impaired renal function, the dose of cefepime should be adjusted to compensate for the slower rate of renal elimination.

The recommended initial dose of cefepime in patients with mild to moderate renal impairment should be the same as in patients with normal renal function. The recommended maintenance dosages of cefepime in adult patients with renal insufficiency are presented in Table 6.

When only a serum creatinine measurement is available, the following formula (Cockcroft and Gault equation) may be used to estimate creatinine clearance. The serum creatinine should represent a steady state of renal function:

Males: Creatinine clearance (mL/min) =
$$\frac{\text{weight (kg)} \times (140 - \text{age})}{72 \times \text{serum creatinine (mg/dL)}}$$

Females: 0.85 x value calculated using formula for males

Table 6: Maintenance Dosing Schedule in Adult Patients with Renal Impairment*

Creatinine clearance (mL/min)	Recommended Maintenance Dosage			
> 50	(Usual dose, no adjustment necessary)			
	2g q8h	2g q12h	1g q12h	500mg q12h
30 - 50	2g q12h	2g q24h	1g q24h	500mg q24h
11 - 29	2g q24h	1g q24h	500mg q24h	500mg q24h
≤ 10	1g q24h	500mg q24h	250mg q24h	250mg q24h
Haemodialysis*	500mg q24h	500mg q24h	500mg q24h	500mg q24h

* Pharmacokinetic modeling indicates that reduced dosing for these patients is necessary. Patients receiving cefepime who are undergoing concomitant haemodialysis should be dosed as follows: 1 gram loading dose on the first day of cefepime therapy and 500 mg per day thereafter. On dialysis days, cefepime should be administered following dialysis. Whenever possible cefepime should be administered at the same time each day.

Dialysis Patients

In patients undergoing haemodialysis, approximately 68% of the total amount of cefepime present in the body at the start of dialysis will be removed during a 3 hour dialysis period. In patients undergoing continuous ambulatory peritoneal dialysis, cefepime may be administered at normally recommended doses, i.e.: 500mg, 1g or 2g, depending on infection severity, but at a dosage interval of every 48 hours.

Children with Impaired Renal Function

Since urinary excretion is the primary route of elimination of cefepime in paediatric patients (see PHARMACOLOGY), an adjustment of the dosage of cefepime should also be considered in patients < 12 years of age with renal impairment.

A dose of 50 mg/kg in patients aged 2 months up to 12 years, and a dose of 30 mg/kg in patients aged 1 month up to 2 months, are comparable to a dose of 2g in an adult. As recommended in Table 6 above, the same increase in interval between doses and/or reduction in dose should be used. When only serum creatinine is available, creatinine clearance may be estimated using either of the following methods:

Creatinine clearance (mL/min/1.73m²) =
$$\frac{0.55 \times \text{height (centimetres)}}{\text{serum creatinine (mg/dL)}}$$

or

Creatinine clearance (mL/min/1.73m²) =
$$\frac{0.52 \times \text{height (centimetres)}}{\text{serum creatinine (mg/dL)}} - 3.6$$

Hepatic Impairment

No dosage adjustment is necessary for patients with impaired hepatic function

Preparation of Solutions and Administration

Cefepime-AFT powder is to be constituted using the volumes of diluent shown in Table 7; the diluents to be used are identified following this table.

Table 7: Preparations of Solutions of Cefepime-AFT

	Amount of diluent to be added (mL)	Approximate available volume (mL)	*Approximate cefepime concentration (mg/mL)
<u>Intravenous</u>			
500mg	5	5.7	90
1g	10	11.4	90
2g	10	12.8	160
<u>Intramuscular</u>			
500mg	1.5	2.2	230
1g	3.0	4.4	230

* **Note:** Reconstitution of Cefepime-AFT in a volume of diluent other than those included in this table will not produce a linear change in concentration

Note: Not all presentations maybe available locally

Intravenous (IV) administration:

The IV route of administration is preferable for patients with severe or life-threatening infections, particularly if the possibility of shock is present. For direct IV administration, Cefepime-AFT should be constituted with Sterile Water for Injection, 5% Dextrose Injection, or 0.9% Sodium Chloride using the diluent volumes shown in Table 7. The resulting solution should be injected directly into the vein over a period of 3 to 5 minutes or injected into the tubing of an administration set while the patient is receiving a compatible IV fluid (see COMPATIBILITY AND STABILITY).

For IV infusion, the 500mg, 1g, or 2g vial of Cefepime-AFT should be constituted as noted above for direct IV administration, then add the appropriate quantity of the resulting solution to an IV container with one of the compatible IV fluids identified under COMPATIBILITY AND STABILITY. Alternatively, the 1g or 2g piggyback (100-mL sized) bottle should be constituted with 50 or 100mL (see Table 7) of compatible IV fluid listed in the COMPATIBILITY AND STABILITY section. The resulting solution should be administered over a period of approximately 30 minutes.

Intramuscular (IM) administration:

Cefepime-AFT should be constituted with one of the following diluents using the volumes shown in Table 7: Sterile Water for injection, 0.9% Sodium Chloride Injection, 5% Dextrose Injection, or Bacteriostatic Water for injection with Parabens or Benzyl Alcohol, then administered by deep IM injection into a large muscle mass (such as the upper outer quadrant of the gluteus maximus). In a pharmacokinetic study, doses up to 1g/vial (volumes < 3.1ml) were administered at single injection sites: the maximum IM dose (2g/6.2ml) was administered at two injection sites. Although Cefepime-AFT can be constituted with 0.5% or 1.0% Lidocaine Hydrochloride, it is usually not necessary because cefepime causes little or no pain upon IM administration.

Compatibility and Stability

Intravenous

Cefepime-AFT is compatible at concentration between 1 and 40 mg/mL with the following I.V. infusion fluids: 0.9% Sodium Chloride, 5% Glucose Injection, M/6 Sodium Lactate Injection, 5% Glucose and 0.9% Sodium Chloride Injection, Lactated Ringers and 5% Glucose Injection.

Cefepime in 0.9% Sodium Chloride or 5% Glucose Injection is compatible when admixed with heparin (10 or 50 units/mL).

Cefepime at a concentration of 40 mg/mL in 0.9% Sodium Chloride or 5% Glucose Injection was found to be compatible with amikacin 6 mg/mL.

Intramuscular

Cefepime-AFT should be reconstituted with the following diluents: Sterile Water for Injections, 0.9% Sodium Chloride, 5% Glucose Injection or 0.5% or 1% lignocaine hydrochloride.

For Both Routes of Administration

Cefepime-AFT should be reconstituted immediately before use and used as soon as practicable after reconstitution, any residue being discarded. If there is any delay in the use of reconstituted Cefepime-AFT it should be stored at 2 – 8 °C for a maximum of 24 hours.

Solutions of cefepime, like those of most beta-lactam antibiotics, should not be added to solutions of metronidazole, vancomycin, gentamicin, tobramycin sulphate or netilmicin sulphate because of physical or chemical incompatibility. However, if concurrent therapy with cefepime is indicated, each of these antibiotics can be administered separately to the same patient.

Note: Parenteral drugs should be inspected visually for particulate matter before administration and not used if particulate matter is present.

As with other cephalosporins, the colour of reconstituted cefepime-AFT may darken on storage. However product potency is not adversely affected.

Reconstituted solutions should be protected from light.

CONTRAINDICATIONS

Cefepime is contraindicated in patients who have shown immediate hypersensitivity reactions to any component of the formulation (including L-arginine), the cephalosporin class of antibiotics, penicillins or other beta-lactam antibiotics.

WARNINGS AND PRECAUTIONS

In patients with impaired renal function, such as reduction of urinary output because of renal insufficiency (creatinine clearance $\leq$ 50 mL/min) or other conditions that may compromise renal function, the dosage of cefepime should be adjusted to compensate for the slower rate of renal elimination. Because high and prolonged serum antibiotic concentrations can occur from usual doses in patients with renal insufficiency or other conditions that may compromise renal function, the maintenance dosage should be reduced when cefepime is administered to such patients. Continued dosage should be determined by degree of renal impairment, severity of infection, and susceptibility of the causative organisms (see DOSAGE and ADMINISTRATION and PHARMACOLOGY). The following serious adverse events have been reported: reversible encephalopathy (disturbance of consciousness

including confusion, hallucinations, stupor and coma), myclonus, seizures (including nonconclusive status epilepticus), and/or renal failure (see ADVERSE REACTIONS). Most cases occurred in patients with renal impairment who received doses of cefepime that exceeded recommendations. In general, symptoms of neurotoxicity resolved after discontinuation of cefepime and/or after haemodialysis however, some cases included a fatal outcome.

Renal function should be monitored carefully if drugs with nephrotoxic potential, such as aminoglycosides and potent diuretics, are administered with cefepime.

Before therapy with cefepime is instituted, careful inquiry should be made to determine whether the patient has had previous immediate hypersensitivity reactions to cefepime, cephalosporins, penicillins, or other beta-lactam antibiotics. Antibiotics should be administered with caution to any patient who has demonstrated some form of allergy, particularly to drugs. If an allergic reaction to Cefepime-AFT occurs, discontinue the drug and treat the patient appropriately. Serious immediate hypersensitivity reactions may require adrenalin and other supportive therapy.

Pseudomembranous colitis has been reported with virtually all broad-spectrum antibiotics including cefepime; therefore, it is important to consider this diagnosis in patients who develop diarrhoea in association with the use of antibiotics. Treatment with broad-spectrum antibiotics alters the normal flora of the colon and may permit overgrowth of clostridia. Studies indicate that a toxin produced by *Clostridium difficile* is a primary cause of antibiotic-associated colitis. Mild cases of pseudomembranous colitis may respond to drug discontinuation alone. In moderate to severe cases, management should include fluid, electrolyte and protein supplementation. When colitis does not improve after drug discontinuation or when it is severe, it should be treated with an antibiotic clinically effective against *Clostridium difficile*. Other causes of colitis should also be considered.

In patients (adult and paediatric) at high risk for severe infection (including patients with a history of recent bone marrow transplantation, with hypotension at presentation, with an underlying haematologic malignancy, or with severe or prolonged neutropenia), antimicrobial monotherapy may not be appropriate. Insufficient data exist to support the efficacy of cefepime monotherapy in such patients.

As with other antibiotics, prolonged use of cefepime may result in overgrowth of non-susceptible organisms. Should super-infection occur during therapy, appropriate measures should be taken.

Cefepime should be prescribed with caution in individuals with a history of gastrointestinal disease, particularly colitis.

If neutropenia occurs as a result of prolonged therapy, cefepime should be discontinued and alternative antibiotic therapy used.

Paediatric Use

Experience with the use of cefepime in paediatric patients aged less than 2 months is limited. Safety and effectiveness in paediatric patients below the age of 2 months have not been established. Therefore the administration of cefepime to patients less than 2 months of age is not recommended.

Geriatric Use

When geriatric patients (65 years or older) received the usual recommended adult dose, clinical efficacy and safety were comparable to clinical efficacy and safety in non-geriatric adult patients unless the patients had renal insufficiency. There was a modest prolongation in elimination half-life and lower renal

clearance values compared to those seen in younger persons. Dosage adjustments are recommended if renal function is compromised (see DOSAGE and ADMINISTRATION).

Cefepime is known to be substantially excreted by the kidney and the risk of toxic reactions to this drug may be greater in patients with impaired renal function. Because elderly patients are more likely to have decreased renal function, care should be taken in dose selection and renal function should be monitored (see PRECAUTIONS, ADVERSE REACTIONS and PHARMACOLOGY).

Serious adverse events, including encephalopathy (disturbance of consciousness including confusion, hallucinations, stupor and coma), myclonus, seizures (including nonconclusive status epilepticus), and/or renal failure have occurred in geriatric patients with renal insufficiency given the usual dose of cefepime (see PRECAUTIONS and ADVERSE REACTIONS).

Carcinogenesis, Mutagenesis, and Impairment of Fertility

Although no long-term studies in animals have been performed to evaluate carcinogenic potential, a battery of in vitro and in vivo tests for genotoxicity have been conducted. The overall conclusion of this testing is that cefepime is not genotoxic.

Standard tests to assess fertility in rats show no impairment of fertility at exposure levels nearly two-fold higher than the calculated maximal daily human exposure.

Driving/Operating Machinery

The effect of cefepime on driving and operating machinery has not been studied. However, possible adverse reactions like altered state of consciousness, dizziness, confusional state or hallucinations may alter the ability to drive and use machines (see WARNINGS, ADVERSE EVENTS, Post-marketing Experience and OVERDOSAGE)

DRUG INTERACTIONS

The pharmacokinetics of cefepime and amikacin when given concurrently to patients with normal renal function do not appear to be altered. Concomitant administration of probenecid prolongs the renal tubular secretion of cephalosporins resulting in high serum concentrations of these antibiotics.

Concomitant treatment with bacteriostatic antibiotics may interfere with the action of betalactam antibiotics.

Cephalosporin antibiotics may produce a false-positive reaction for glucose in the urine with copper reduction tests (Benedict's or Fehling's solution or with Clinitest tablets), but not with enzyme-based tests (glucose oxidase) for glycosuria. Therefore, it is recommended that glucose tests based on enzymatic glucose oxidase reactions be used.

INCOMPATIBILITIES

Renal functions should be monitored carefully if drugs with nephrotoxic potential, such as aminoglycosides and potent diuretics, are administered with cefepime.

Nephrotoxicity has been reported following concomitant administration of other antibiotics with aminoglycoside antibiotics or potent diuretics such as frusemide.

USE IN PREGNANCY AND LACTATION

Use in Pregnancy

Category B1

Reproduction studies performed in mice and rats showed no evidence of impaired fertility or harm to the foetus at dose levels equivalent to (mouse) or slightly greater (rat) than the maximum human daily dose when the daily doses are compared to those in man on a mg/m² basis. There are, however, no adequate and well-controlled studies in pregnant women. Because animal reproduction studies are not always predictive of human response, this drug should be used during pregnancy only if clearly needed.

Use in Lactation

Cefepime is excreted in human breast milk in very low concentrations. Although less than 0.01% of a 1g/vial I.V. dose is excreted in milk, caution should be used when cefepime is administered to a nursing woman.

ADVERSE EFFECTS

Cefepime is generally well tolerated. The most common adverse events were gastrointestinal symptoms and hypersensitivity reactions, presented in Table 8 below.

Table 8: Adverse Events Reported

Hypersensitivity	Anaphylaxis		Pruritus, urticarial	Rash
Gastrointestinal		Abdominal pain, constipation, unspecified moniliasis	Nausea, vomiting, oral moniliasis, colitis (including <i>Pseudomonas colitis</i>)	Diarrhoea
Central nervous system	Seizures	Dizziness, paraesthesia, taste perversion	Headache	
Other		Vasodilation, dyspnoea, genital pruritis, chills, inflammation at the site of IV infusion	Fever, vaginitis, erythema	Local reactions at the site of IV infusion including phlebitis, inflammation or pain at the site of IM injection

Laboratory test abnormalities that developed in patients with normal baseline values were transient. Those that occurred were: elevations in alanine aminotransferase, aspartate aminotransferase, alkaline phosphatase, total bilirubin, anaemia, eosinophilia, prolonged prothrombin time, partial prothrombin time, and positive Coombs' test without haemolysis. Transient elevations of serum urea, and/or serum creatinine and transient thrombocytopenia were also observed. Transient leucopenia and neutropenia were also seen.

Post-marketing Experience

During post-marketing experience, encephalopathy (disturbance of consciousness including confusion, hallucinations, stupor and coma), seizures, myoclonus and/or renal failure have been reported. Most

cases occurred in patients with renal impairment who received doses of cefepime that exceeded recommendations (see also PRECAUTIONS).

Anaphylaxis including anaphylactic shock, transient leucopenia, neutropenia, agranulocytosis and thrombocytopenia have been reported rarely.

Because of the uncontrolled nature of these spontaneous reports, a causal relationship to Cefepime has not been determined.

The following adverse reactions and altered laboratory tests have been reported for cephalosporin-class antibiotics: urticaria, Stevens-Johnson syndrome, erythema multiforme, toxic epidermal necrolysis, renal dysfunction, toxic nephropathy, aplastic anaemia, haemolytic anaemia, haemorrhage, hepatic dysfunction including cholestasis, and false positive tests for urinary glucose.

Paediatrics

The safety profile of Cefepime in infants and children is similar to that seen in adults. The most frequently reported adverse event considered related to cefepime was rash.

OVERDOSAGE

In case of severe overdosage, especially in patients with compromised renal function, dialysis will aid in the removal of cefepime from the body; peritoneal dialysis is of no value. Accidental overdosing has occurred when large doses were given to patients with impaired renal function (see DOSAGE AND ADMINISTRATION, PRECAUTIONS and ADVERSE REACTIONS). Symptoms of overdosage include encephalopathy (disturbance of consciousness including confusion, hallucinations, stupor, and coma), myoclonus, seizures and neuromuscular excitability.

DOSAGE FORMS AND PACKAGING AVAILABLE

Cefepime-AFT is a sterile dry mixture of cefepime hydrochloride and L-arginine. It is supplied in vials containing cefepime hydrochloride equivalent to cefepime 1g and 2g.

Cefepime-AFT is available in packs of 1, 5 and 10 vials.

Note: Not all presentations may be available locally

STORAGE AND SHELF LIFE

Shelf life: 2 years or 24 months from the date of manufacture

Store in the dry state in original carton.

Store at or below 30 °C. Protect from light.

To avoid the risk of microbial contamination, reconstituted Cefepime-AFT should be administered as soon as possible after reconstitution. If storage is needed, store at 2 – 8°C for not more than 24 hours (See Compatibility and Stability)

PRODUCT OWNER

New Zealand Office:
AFT Pharmaceuticals Ltd.

PO Box 33-203, Takapuna, Auckland 0740
Level 1, Nilsen Building, 129 Hurstmere Road, Takapuna, Auckland 0622, New Zealand

Australian office:
AFT Pharmaceuticals Pty. Ltd.
PO Box 748 North Ryde NSW 1670
Suite 301, 113 Wicks Road, North Ryde, Sydney, NSW 2113, Australia

PRODUCT REGISTRATION HOLDER

For Singapore:
DKSH Singapore Pte Ltd
24 Penjuru Road, #03-02, Singapore 609128

For Malaysia:
AFT Pharmaceuticals (S.E. Asia) Sdn. Bhd.
A-29-3, Level 29, Menara UOA Bangsar, No.5, Jalan Bangsar Utama 1,
59000 Kuala Lumpur, Malaysia

MANUFACTURER

Qilu Pharmaceutical Co., Ltd.
No. 317, Xinluo Road, High-Tech Zone, Jinan, Shandong Province 250101, P.R. China

Registration Numbers:

1g vial SIN14658P, MALxxxxxxxACZ

Date of revision: February 2018